WeResence.

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WeResence — Product Requirement Document

Sovereign Dual-Transport Hybrid Messenger & Private AI Ecosystem • Author: Piyush Singh

> **Executive Summary:** WeResence is a privacy-first communication platform that guarantees message continuity through a dual-engine architecture: WebSockets over TLS 1.3 when online, and stealth port-directed cellular Data SMS on Port 8901 when offline. It integrates a dedicated Resence AI assistant, hardware-enforced anti-screenshot protection, mutual private vaults, and zero native call log traces.

1. Problem Statement & Market Opportunity

Modern messengers (WhatsApp, Signal, Telegram) rely 100% on active data packets. In cellular blindspots (underground transit, campus basements, rural transit, high-density gatherings), users experience complete communication blackouts. Simultaneously, users face over-the-shoulder snooping, device-level screenshot risks, and cloud data retention vulnerabilities. WeResence bridges this gap with seamless cellular-level fallback and military-grade hardware isolation.

2. Target Personas

Persona	Profile	Primary Requirement
Piyush (Sovereign Founder)	Tech innovator & student	Absolute zero-leak privacy, AI companion integration, zero telecom surveillance.
Campus Students & Youth	Gen-Z daily communicators	Linear/Apple minimal aesthetics, shared secret photo vault with friends, anti-screenshot.
Transit Professionals	Frequent travelers	Reliable messaging across low-reception railway/highway transit dead zones.

3. Core Product Functional Requirements

3.1 Dual-Engine Transport Routing

- **Automatic Switching:** Real-time network monitor transitions between Internet and Port 8901 SMS in under 20ms without manual intervention.

- **1-on-1 SMS Isolation:** Offline SMS fallback is strictly restricted to 1-on-1 direct conversations. Group and community chats remain 100% online to protect carrier SMS limits.
- **Inbox Stealth:** Binary SMS messages delivered to Port 8901 are captured with priority 999, completely invisible to Google Messages and Samsung Messages.
- **Carrier Quota Safeguard:** Displays an inline non-intrusive meter (e.g. 84/100 today). When 100 SMS are reached, messages queue safely in the local Outbox.

3.2 Built-In Resence AI (@ResenceAI)

- **Multimodal Companion:** Pinned at the top of the chat list for coding, research, and ideation.
- **Strict Offline Freeze:** When data disconnects, Resence AI turns 100% grayscale and disables input to conserve battery. Zero background queueing.

3.3 Mutual Private Vault Gallery (1-on-1)

- **Header Pathway:** Accessible by tapping the contact header and scrolling to the bottom.
- **Independent Passwords:** Person A and Person B set their own custom passwords. Neither knows the other's secret PIN.
- **Uploader-Only Delete Governance:** The original uploader can delete their media anytime. The recipient has view-only permissions, preventing malicious wipeouts.

3.4 Sovereign Voice Calling

- **VoIP Audio:** End-to-end encrypted voice calls over internet/Wi-Fi with zero carrier billing.
- **Zero Call Logs:** System dialer call logs are automatically scrubbed upon call termination.

4. UX & Non-Functional Guidelines

- **Zero Jargon:** No technical acronyms in user UI. Clear, plain language throughout.
- **Inline Pills:** All alerts appear as subtle, rounded 10px inline pills (WhatsApp style).
- **Memory Budget:** Idle RAM < 40MB | Active RAM < 65MB on 3GB/4GB budget devices.